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St. Joseph's College of Arts & Science (Autonomous)
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M.Sc Degree Examination – APRIL 2021

EPCH912S – PHYSICAL METHODS IN ORGANIC CHEMISTRY - III

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Among the following which transition requires least energy?
   a) n→π* b)π→π* c)σ→σ* d)n→σ*

2. The number of possible Uv absorption bands for benzene is
   a) 1 b) 2 c) 3 d) 4

3. Carbonyl group shows strong absorption in the region
   a) 1100-1600cm^{-1} b) 1900-1600cm^{-1} c) 3300-2100cm^{-1} d) All the above

4. Aromatic Ar-H stretching frequency will be in the region
   a) 3050-3000cm^{-1} b) 3000-2900cm^{-1} c) 2050-2000cm^{-1} d) 1900- 1600cm^{-1}

5. The correct order for the basic features of mass spectrometer is
   a) Acceleration, deflection, detection, ionization
   b) Ionization, Acceleration, deflection, detection
   c) Acceleration, ionization, deflection, detection
   d) Acceleration, deflection, ionization, detection

6. Major peaks in mass spectrum of but-2-yne appear at m/z=54,39 and 27. Which of the following is inconsistent with these data?
   a) But-2-yne fragments by loss of two methyl groups
   b) But-2-yne fragments by cleavage of C-C bond
   c) But-2-yne fragments by loss of a methyl group
   d) The highest mass peak corresponds to $[C_4 H_6]^+$

7. The statement about TMS which is not Correct is,
   a) It gives a single peak b) It has one kind of H
   c) It gives high chemical shift values d) It's hydrogens are more shielded

8. The number of signals for the compound A-CH_2CH_2-B is
   a) Two triplets b) Two Singlet c) One Singlet d) One triplet

9. The spin quantum number for ^{13}C is
   a) Zero b) 1/2 c) One d) All the above

10. ^{13}CMR can detect
    a) Total number of protons b) Total number of Carbons
    c) Presence of carbonyl group d) All the above

II. Fill up the blanks

11. λ_{max} for 2,4-hexadiene is _____.

12. The total number of Vibrational degrees of freedom for benzene is _____.

13. Molecular weight of a molecule can be determined in mass spectroscopy from _____.

14. NMR Signals for CH_3 CH_2 Br is _____.

15. A compound C_2H_6O gives one proton singlet and a ^{13}C NMR quartet, structure of compound is _____.

III. Match the following

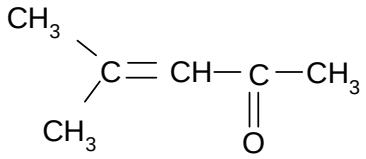
16. K-band	a. 800-600cm^{-1}
17. C-cl stretching	b. Carbon skeleton
18. Mc Lafferty rearrangement	c. TMS
19. PMR	d. $\pi \rightarrow \pi^*$ transition
20. ^{13}C NMR	e. Fragmentation Pattern

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. State Beer-Lambert law.

22. Calculate λ_{max} for



23. Write the selection rule for IR spectroscopy.

24. Explain Fermi resonance using CO_2 molecule.

25. Define base peak and molecular ion peak.

26. Write the M+1 and M+2 peaks of benzene.

27. Explain Mc- Lafferty rearrangement.

28. What is NOE?

29. Why is TMS used as a reference in NMR?

30. How many NMR Signals are Possible for Cyclohexane and 2-chloropropene.

31. What is SEFT spectra?

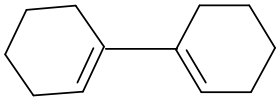
32. What is number of Peaks for Phenyl acetic acid in ^{13}CNMR?

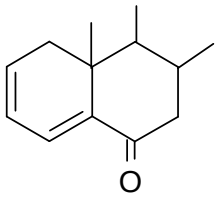
SECTION- C (5X7=35)

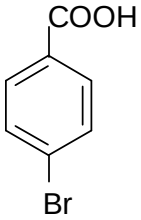
Answer any **FIVE** Questions

33. Write a note different types of transitions in UV spectroscopy.

34. Calculate λ_{max} for

a) 

b) 

c) 

35. Discuss how hydrogen bonding can be Predicted by IR spectroscopy.

36. A compound C_8 H_6 show m/e values 102, 76 and 50 in mass spectrum. Predict the compound.

37. Define chemical shift in NMR. Explain the factors influencing chemical shift.

38. An organic compound with molecular mass 120 absorbs in UV spectrum at 268nm, Σ_{max} 480. In IR spectrum, medium absorption bands are formed at 3067-2907, 1608 and 1473cm^{-1} respectively. The NMR spectrum shows i) 3.21τ Singlet (10.4 squares) ii) 7.74τ Singlet (31 squares). Predict the structure of the compound.

39. Using proton-decoupled ^{13}CNMR spectra, distinguish the following isomeric compounds.
    a) 2-Pentanone and 3-Pentanone
    b) 1-Propanol and 2- propanol

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